ENEL 573 HW Assignment 4 (Fall 2026)

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Problem 1

Class B: | 10 | NET ID (14 bit) | SUBNET ID (n bit) | HOST ID (16-n bit) |

2^n ≥ 950 → 2^9 = 512 < 950, 2^10 = 1024 ≥ 950 → n = 10

HOST ID = 16 - 10 = 6 bits

Mask: 11111111 11111111 11111111 11000000

Ans: Subnet mask = 255.255.255.192 (/26)

subnets = 2^10 = 1024, addresses/subnet = 2^6 = 64

max hosts/subnet = 2^6 - 2 = 62

e.g. network 150.100.0.0, X = 10-bit subnet ID:

Start (subnet addr): 10010110 01100100 XXXXXXXX XX000000

Last (broadcast) : 10010110 01100100 XXXXXXXX XX111111

X = 0000000000 to 1111111111

Subnet	X	Subnet address	Host range	Broadcast
#0	0000000000	150.100.0.0/26	150.100.0.1 to 150.100.0.62	150.100.0.63
#1	0000000001	150.100.0.64/26	150.100.0.65 to 150.100.0.126	150.100.0.127
#2	0000000010	150.100.0.128/26	150.100.0.129 to 150.100.0.190	150.100.0.191
#3	0000000011	150.100.0.192/26	150.100.0.193 to 150.100.0.254	150.100.0.255
…	...	…	…	…
#949	1110110101	150.100.237.64/26	150.100.237.65 to 150.100.237.126	150.100.237.127
#1023	1111111111	150.100.255.192/26	150.100.255.193 to 150.100.255.254	150.100.255.255

Problem 2

120 → class A (/8). Mask 255.255.192.0 = /18 → 10 subnet bits, 14 host bits

IP : 01111000 00001110 01111010 00010000

Mask: 11111111 11111111 11000000 00000000

AND : 01111000 00001110 01000000 00000000 = 120.14.64.0

Ans: Subnet address = 120.14.64.0/18

Host id = IP AND (NOT mask) = 00111010 00010000 (last 14 bits)

Ans: Host id = 0.0.58.16

Problem 3

Data = 1300 - 20 = 1280 bytes

Max data/fragment = 500 - 20 = 480 bytes (480/8 = 60 ✓)

1280 = 480 + 480 + 320

Frag	length	ID	flag	offset
1	=500	=X	=1	=0
2	=500	=X	=1	=480/8 = 60
3	=340	=X	=0	=960/8 = 120

Ans: Option 2 (offsets 0, 480/8, 960/8). Option 1 ignores the 20-byte header; option 3 subtracts it twice (460 is not a multiple of 8).

Problem 4

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Datagram OK only if all 10 fragments arrive.  p = 0.01
(a) P = (1 - p)^10 = (0.99)^10
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Ans: (a) P ≈ 0.9044

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(b) fragment lost only if both copies lost: p^2 = 0.0001
    P = (1 - p^2)^10 = (0.9999)^10
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Ans: (b) P ≈ 0.9990

Problem 5

Frames to B's MAC address: Yes, C's adapter processes them. On a broadcast LAN every frame reaches every adapter, and C's adapter checks the destination MAC. No, it does not pass the IP datagrams to C's network layer: the destination MAC is B's, not C's, so C's adapter discards the frame.

Frames to broadcast address FF-FF-FF-FF-FF-FF (ARP query): Yes, C's adapter processes them, and yes, it passes the contents up to C's network layer, since every adapter accepts the broadcast address. C then sees the target IP is B's, not its own, so it ignores the ARP query and does not reply (only B replies, by unicast to A).